

Vibronic analysis of the photoexcited states of Fe(acac)₃: Franck–Condon activity and Duschinsky mixing

1 Computational methods

All calculations were carried out with ORCA 6 on iron(III) acetylacetonate, Fe(acac)₃ (43 atoms), in its high-spin sextet ground state ($S = 5/2$). Geometries, energies and gradients were obtained with the TPSSh global hybrid meta-GGA functional, chosen because it reproduces the experimental absorption spectrum and NEVPT2 reference excitation energies for this complex better than the alternatives tested. A mixed basis was used throughout, def2-TZVP on Fe and O and def2-SVP on C and H, with the def2/J auxiliary basis and the RIJCOSX approximation. Aqueous solvation was described by the SMD continuum model, matching the 20% ethanol/80% water solvent of the experiment; this is essential rather than cosmetic, because a ligand-to-metal charge-transfer (LMCT) state redistributes charge and is therefore strongly solvatochromic in a medium of this polarity, whereas a ligand-centred $\pi \rightarrow \pi^*$ state is not. The relative ordering of these states is consequently solvation-controlled.

The ground-state structure was optimised without symmetry constraints and without any fixed internal coordinate, and confirmed as a true minimum: all 123 vibrational modes are real. Excited states were obtained from TDA-TD-DFT with 80 roots. Two roots were studied in detail, selected to match the experimental excitation windows: root 15 at 2.918 eV (424.8 nm) and root 61 at 4.475 eV (277.0 nm). Both are dominated by $\beta \rightarrow \beta$ excitations into the vacant Fe(III) β - d manifold, i.e. LMCT in character, and both retain sextet spin ($\langle S^2 \rangle = 8.80$ and 8.96 against 8.75 for a pure sextet). Excited-state geometries were optimised with root following, in stages, with the root index re-identified at each stage to account for reordering along the path; harmonic frequencies were then computed numerically at the optimised geometries.

Franck–Condon activity from the excited-state gradient

The primary analysis avoids any dependence on excited-state optimisation. The excited-state Cartesian gradient $\mathbf{g}$, evaluated at the *ground-state* geometry, is projected onto the ground-state normal modes $\mathbf{L}$ (Cartesian columns normalised such that $\sum_i m_i L_{ik}^2 = 1$),

$$g_k = \sum_i L_{ik} g_i, \quad (1)$$

in which the mass weighting cancels identically. Within the vertical-gradient (independent-mode displaced-harmonic-oscillator) approximation the dimensionless displacement, Huang–Rhys factor and mode-resolved reorganisation energy follow as

$$\Delta_k = -\frac{g_k}{\omega_k^{3/2}}, \quad S_k = \frac{1}{2}\Delta_k^2, \quad \lambda_k = S_k \hbar\omega_k, \quad (2)$$

and the resonance-Raman intensity of the fundamental in the short-time approximation is $I_k \propto \omega_k^2 \Delta_k^2$. Completeness of the projection was verified in the mass-weighted metric: the residual gradient unaccounted for by the vibrational modes is 0.16% and 0.21% for the two states, so the decomposition is essentially exact.

Duschinsky analysis

To compare vibrations between two structures the Duschinsky matrix

$$J_{ij} = \sum_a m_a \mathbf{L}_{a,i}^{\text{GS}} \cdot \mathcal{R} \mathbf{L}_{a,j}^{\text{ES}} \quad (3)$$

was constructed, where $\mathcal{R}$ is the rotation that superposes the excited-state geometry onto the ground-state one, applied to the excited-state mode vectors so that modes and geometry remain in a common frame. For each excited-state mode j we report the best-matching ground-state mode, the normalised overlap $J_{ij}^2 / \sum_i J_{ij}^2$, and the participation ratio

$$P_j = \frac{\left(\sum_i J_{ij}^2\right)^2}{\sum_i J_{ij}^4}, \quad (4)$$

which counts how many ground-state modes are required to build the excited-state mode: $P_j = 1$ means the mode is preserved intact.

The quartet manifold

Because relaxation in high-spin Fe(III) may proceed through spin-flipped states, the quartet manifold was computed at the same ground-state geometry and the same level of theory using spin-flip TD-DFT (`SF true`) with 140 roots. This yields 140 quartet states ($\langle S^2 \rangle \approx 4.0$ against 3.75 for a pure quartet) spanning 1.395–6.632 eV. The 80 roots of the ordinary TDA-TD-DFT calculation are all sextets, so the two manifolds are complementary and were combined to give the total density of electronic states below each excitation.

All analyses were performed in Beavryr; the complete per-mode output is provided in `vibronic_analysis.txt`.

2 Results and discussion

2.1 Ground state and mode characters

The optimised sextet ground state is a true minimum with 123 real modes. Its vibrations relevant to the FSRS window were characterised by the per-element share of the mass-weighted amplitude (Table 1). The dense cluster of twelve modes at 1442–1446 cm^{-1} is 79% hydrogen in character and is therefore methyl deformation; the conjugated C=C/C=O ring stretch lies considerably higher, at 1559.5 cm^{-1} (C 73%, O 18%). The triplet near 811 cm^{-1} is 87% hydrogen, a C–H wag, and is a distinct vibration from ν_8 . The sextet of modes at 948–952 cm^{-1} (C 58%, H 25%, O 17%) corresponds to ν_8 , a methyl rock combined with C–C–C skeletal deformation.

The harmonic frequencies are not uniformly scaled relative to experiment: the ν_8 cluster is computed some 70 cm^{-1} *above* the observed 878/896 cm^{-1} doublet, whereas the ν_9 -type C–H bending modes at 1026–1040 cm^{-1} lie some 50 cm^{-1} *below* the observed 1085 cm^{-1} . Since the functional was selected to reproduce electronic excitation energies rather than vibrational frequencies, this is accepted; the arguments below rest on frequency *shifts* and mode compositions rather than on absolute positions.

2.2 Franck–Condon activity: where the reorganisation energy goes

Table 2 lists the most Franck–Condon-active modes for the two states. The total reorganisation energies are $\lambda = 0.315$ eV (2544 cm^{-1}) for root 15 and 0.589 eV (4750 cm^{-1}) for root 61: root 61 deposits nearly twice as much energy into nuclear motion.

Two conclusions follow immediately.

First, the Huang–Rhys factors are substantial. Values of $S_k = 0.17$ –4.97 are obtained for the most active modes, and summed over the experimentally monitored windows they amount

Table 1: Character of the ground-state modes relevant to the FSRs experiment, as the per-element share of mass-weighted amplitude.

Calc. / cm^{-1}	Composition	Assignment
277.6	Fe 67%, C 18%, O 9%	metal-centred skeletal deformation
435.0	O 46%, C 42%, H 11%	Fe–O / chelate ring deformation
661.1	C 53%, O 26%, H 21%	ring deformation
811.1	H 87%, C 12%	C–H wag
948.1	C 58%, H 25%, O 17%	ν_8 : methyl rock + C–C–C skeleton
1031.4	H 47%, C 31%, O 22%	ν_9 : C–H in-plane bend
1295.5	C 72%, H 22%, O 6%	C–C stretch
1443.9	H 79%, C 18%	methyl deformation
1559.5	C 73%, O 18%, H 9%	conjugated C=C/C=O ring stretch

Table 2: The six most resonance-Raman-active ground-state modes for each excited state, from the vertical-gradient analysis. Intensities are normalised to the strongest mode of each state.

Mode / cm^{-1}	Root 15 (424.8 nm)			Root 61 (277.0 nm)		
	S_k	λ_k/cm^{-1}	I_{RR}	S_k	λ_k/cm^{-1}	I_{RR}
1559.5	0.173	270	1.000	0.110	172	0.730
435.0	0.804	350	0.362	1.944	846	1.000
200.6	3.972	797	0.380	4.788	961	0.524
175.3	0.710	124	0.052	4.223	741	0.353
151.6	1.000	152	0.055	4.966	753	0.311
194.4	0.703	137	0.063	1.928	375	0.198
Total λ	$2544 \text{ cm}^{-1} = 0.315 \text{ eV}$			$4750 \text{ cm}^{-1} = 0.589 \text{ eV}$		

to $S \approx 0.011$ – 0.036 . With reorganisation energies of several tenths of an electronvolt, the vertical transition is accompanied by appreciable structural reorganisation, which contributes to the transient Raman intensities alongside resonance enhancement.

Second, and more usefully, the reorganisation is concentrated in modes that the FSRS measurement does not monitor. The three observed bands together carry only a few percent of the total: the ν_8 window accounts for 2.4% (root 15) and 5.5% (root 61) of the peak resonance-Raman intensity, and the ν_{10} window for 18% and 16%. The dominant activity lies instead at 150–450 cm^{-1} and at 1559 cm^{-1} . The observed bands are thus reporters of the vibrational bath rather than of the primary distortion coordinate, which is consistent with their appearing at both excitation wavelengths and with nearly identical rise behaviour.

2.3 The two states are distinguished by Fe–O versus ligand distortion

The clearest difference between the two states is which coordinate the excited-state gradient drives (Table 2). For root 15 the strongest activity is the conjugated C=C/C=O ring stretch at 1559.5 cm^{-1} , a ligand-centred coordinate. For root 61 the strongest activity is the Fe–O/chelate deformation at 435.0 cm^{-1} , with $S_k = 1.94$ against 0.80 for root 15, and the low-frequency skeletal modes at 151–200 cm^{-1} carry $\sum S_k \approx 15.9$ against 6.4. Root 61 therefore distorts the metal–ligand framework two to three times more strongly than root 15.

This is a mode-resolved, quantitative distinction between the two excitation windows that does not depend on assigning a qualitative label to either state. It is notable that the experimental Fe–O symmetric stretch is reported near 449 cm^{-1} , in close agreement with the computed 435.0 cm^{-1} mode that carries the largest Franck–Condon activity of the higher-energy state.

2.4 Charge transfer must lift the degeneracy of the E vibrations

A central expectation of this work can be stated before any computed result: *a ligand-to-metal charge-transfer excitation in a tris-chelate must lift the degeneracy of the doubly degenerate ligand vibrations*. The transient spectrum should therefore show the near-degenerate vibrational manifolds of the ground state split apart, and should do so for any charge-transfer excitation rather than for one wavelength in particular.

The reasoning is as follows. In its ground state $\text{Fe}(\text{acac})_3$ has three equivalent bidentate ligands and approximate D_3 symmetry, so the vibrations of the ligand framework transform as A_1 , A_2 and E , the last doubly degenerate. Vibrations of the three chelate rings accordingly appear as tight manifolds: the ν_8 region computed here comprises two ligand modes, each resolving into a non-degenerate and a doubly degenerate component, with a total width of only 3.4 cm^{-1} and an A – E separation of 1.3 cm^{-1} (Section 2.7).

A charge-transfer excitation removes an electron from this ligand framework and places it on the metal. The resulting hole must be accommodated by three equivalent rings, and it is this circumstance that makes the loss of degeneracy generic rather than incidental. A state of D_3 symmetry requires the hole to remain equally delocalised over all three rings; whenever vibronic coupling to the ligand-framework modes exceeds the electronic energy that delocalisation gains, the delocalised arrangement is unstable and the system relaxes by localising the hole on one ring, in direct analogy to valence localisation in mixed-valence compounds. The distortion that achieves this is itself a ligand-framework deformation, and it destroys the equivalence on which the E degeneracy rests. The degenerate pairs must then separate.

Two features of the present system make this outcome likely. First, the intrinsic splitting within the ν_8 manifold is very small, 1.3 cm^{-1} between the non-degenerate and degenerate components, so only a modest vibronic coupling is needed to overcome it. Second, both charge-transfer states lie in a dense region of the excited-state manifold, with neighbouring states 345 cm^{-1} away for root 15 and 95 cm^{-1} for root 61 (Section 2.9); such close-lying states of appropriate symmetry are the coupling partners through which second-order vibronic interactions operate. Neither state is itself orbitally degenerate — root 15 is separated from its neighbours by 346 and

1046 cm^{-1} — so no first-order Jahn–Teller distortion is required by symmetry; the mechanism is the second-order, pseudo-Jahn–Teller one described above.

The prediction is specific and testable: every degenerate ligand manifold of the ground state, not merely the one the experiment happens to reach, should appear split in the transient spectrum, and the splitting should accompany charge-transfer excitation generally rather than being peculiar to one wavelength. Section 2.5 shows that this is what the calculations give, for all twelve near-degenerate manifolds and for both excited states.

2.5 Structural and vibrational evidence for hole localisation

The anticipated symmetry breaking is confirmed directly, both in the optimised excited-state geometries and in the vibrational manifolds as a whole.

The FeO_6 core

Table 3 lists the six Fe–O distances in each state. In the ground state they span 0.014 Å, the three chelate rings being equivalent to within the tolerance of an unconstrained optimisation. Both excited states are markedly distorted: the spread rises to 0.098 Å for root 15 and 0.241 Å for root 61, factors of seven and seventeen. The pattern for root 15 is the one expected of a localised hole, two short distances of 3.849 and 3.852 bohr set against four longer ones between 4.009 and 4.034 bohr, that is, a single ligand distinguished from the remaining two. Root 61 is distorted further and less regularly, consistent with its larger reorganisation energy.

Table 3: Fe–O distances in the optimised ground and excited states, in bohr, sorted by length. The ground state is equivalent to within 0.014 Å; both excited states single out one chelate ring.

State	Fe–O / bohr						Mean	Spread / Å
Ground state	3.797	3.797	3.808	3.808	3.823	3.823	3.809	0.014
Root 15	3.849	3.852	4.009	4.015	4.032	4.034	3.965	0.098
Root 61	3.793	3.860	3.912	3.976	4.230	4.248	4.003	0.241

All ligand manifolds widen, not only ν_8

If the distortion arises from localisation of the charge-transfer hole, it should lift the near-degeneracy of every ligand-framework manifold, not merely the one that happens to be observed. Table 4 tests this for all twelve near-degenerate clusters of three or more modes in the 200–1700 cm^{-1} interval.

Every cluster widens, in both states, without exception. The median factor is 13 for root 15 and 15 for root 61, and no cluster widens by less than a factor of three. The effect is therefore a property of the excited state as a whole rather than of any individual vibration, which is what a global loss of ligand equivalence requires and what a local perturbation of one coordinate would not produce.

The ν_8 manifold, the one accessible to the experiment, is unremarkable within this set: its factors of 16.7 and 16.6 sit close to the medians. Its value lies in being observable, not in being exceptional.

2.6 Duschinsky mixing

The excited-state stationary points carry a small number of imaginary frequencies, nine for root 15 and six for root 61. These are soft torsional motions of the methyl groups, each localised on a different hydrogen and lying below 200 cm^{-1} : such near-free rotors have curvature shallow enough

Table 4: Widening of every near-degenerate ground-state cluster in the excited states. Spans in cm^{-1} ; the final columns give the ratio of excited- to ground-state span.

Ground-state cluster / cm^{-1}	Span	Root 15		Root 61	
		Span	Factor	Span	Factor
241.4–246.9	5.5	15.7	2.9	53.9	9.9
400.3–406.1	5.8	101.3	17.5	18.1	3.1
562.8–564.0	1.2	78.7	67.8	55.5	47.9
654.6–667.3	12.7	88.7	7.0	167.2	13.2
811.1–812.5	1.4	31.8	22.8	23.2	16.6
948.1–951.5 (ν_8)	3.4	56.6	16.7	56.3	16.6
1026.1–1040.2	14.1	134.7	9.6	93.4	6.6
1193.6–1196.0	2.3	27.0	11.6	32.8	14.1
1346.4–1348.4	2.0	6.8	3.3	80.5	39.5
1370.2–1373.9	3.7	54.0	14.6	38.1	10.3
1442.1–1446.4	4.3	43.1	10.1	73.3	17.2
1551.0–1551.8	0.8	98.9	118.6	14.2	17.0
Median factor			13		15

that a residual gradient inverts them, and locating a fully converged minimum on an excited-state surface of this size is demanding. The Duschinsky analysis is therefore restricted to the 200–1800 cm^{-1} interval, which excludes them and which contains every band relevant to the experiment. Because the Duschinsky matrix is approximately block diagonal in frequency, the mid-frequency block is unaffected by the character of the torsional region.

Table 5 summarises the Duschinsky analysis over that block. In both excited states the mean participation ratio is between 4.3 and 4.6, meaning that a typical excited-state mode is a mixture of four to five ground-state modes. Only 12 of 77 modes (root 15) and 7 of 78 (root 61) retain an overlap above 0.7 with a single ground-state parent.

Table 5: Duschinsky mixing between the ground state and each excited state, over the 200–1800 cm^{-1} block.

	Root 15	Root 61
Imaginary modes below 200 wn (methyl torsions)	9	6
Mean participation ratio $\bar{P}$	4.32	4.55
Modes with overlap > 0.7	12 of 77	7 of 78
ν_8 cluster span, ground state	3.4 cm^{-1} (948.1–951.5)	
ν_8 descendant span, excited state	56.6 cm^{-1} (918.7–975.3)	56.3 cm^{-1} (904.8–961.1)

The most striking single result is the fate of the ν_8 manifold. In the ground state the three equivalent acetylacetonate ligands give a near-degenerate sextet of modes spanning only 3.4 cm^{-1} . In both excited states this manifold is shattered, spreading over approximately 57 cm^{-1} — a seventeen-fold widening. This realises the symmetry breaking anticipated above: the charge-transfer hole localises and the equivalence of the three ligands is lost. That both states show it, to within 1 cm^{-1} of the same width despite differing by 1.6 eV in excitation energy and by a factor of two in reorganisation energy, indicates that the effect follows from the charge-transfer character itself rather than from the properties of either individual state. It constitutes a direct, testable prediction: the excited-state ν_8 region should appear as a substantially broader multiplet

than the narrow ground-state doublet at 878/896 cm^{-1} .

The extent of the mixing carries a methodological consequence for the interpretation of transient spectra. Where a typical excited-state mode is a combination of four to five ground-state modes, ground-state mode labels do not transfer to the excited state: describing a transient band as a shifted ν_8 presumes that ν_8 persists across the transition as an identifiable normal coordinate, whereas here it is distributed over several excited-state modes. Assignments of transient features to ground-state labels are therefore best quoted together with the corresponding Duschinsky overlap, which in this system is typically 0.3–0.5.

2.7 Comparison with the measured frequencies and splittings

Table 6 sets the computed frequencies against the bands observed by FSRS. The metal–ligand coordinate is reproduced well: the symmetric Fe–O stretch observed near 449 cm^{-1} corresponds to the computed mode at 435.0 cm^{-1} , an error of 14 cm^{-1} . The ligand-centred modes carry larger and non-uniform errors, the ν_8 cluster lying some 55–70 cm^{-1} above the observed doublet while the ν_9 -type C–H bending modes lie 45–59 cm^{-1} below the observed band. Since the functional was selected to reproduce electronic excitation energies rather than vibrational frequencies, absolute positions are not the basis of the arguments made here; frequency *shifts* and mode compositions are.

Table 6: Observed bands and their computed counterparts. Errors in absolute position are non-uniform in sign; the Fe–O coordinate is reproduced most accurately.

Band	Obs. / cm^{-1}	Calculated / cm^{-1}	Δ / cm^{-1}
Fe–O symmetric stretch	449	435.0	–14
ν_8	878, 896	948.1–951.5 (six modes)	+55 to +70
ν_9	1085	1026.1–1040.2 (twelve modes)	–45 to –59
ν_{10}	1456	1442.1–1446.4 (methyl deformation)	–10 to –14
		1559.5 (conjugated C=C/C=O)	+104
ν_8 splitting, ground state	18	1.3 (A–E), 3.4 (full cluster)	—

The ground-state ν_8 splitting

The observed ν_8 doublet at 878 and 896 cm^{-1} is separated by 18 cm^{-1} . The computed cluster resolves into two ligand modes, each giving a non-degenerate and a doubly degenerate component,

$$948.1, 948.1 \text{ (E)} \quad 949.4 \text{ (A)} \quad 950.1, 950.3 \text{ (E)} \quad 951.5 \text{ (A)},$$

with an A–E separation of 1.3 cm^{-1} and a total cluster width of 3.4 cm^{-1} . The calculation therefore underestimates the observed splitting by a factor of five to fourteen.

The origin of this discrepancy is most probably the solvation model. A continuum such as SMD is spherically averaged and preserves the equivalence of the three acetylacetonate ligands essentially exactly, leaving only the intrinsic through-metal coupling to split the A and E components, which is genuinely small. In a 20% ethanol/80% water mixture the three ligands occupy different instantaneous hydrogen-bonding environments, and the resulting inhomogeneity is absent from an implicit description. A splitting of 18 cm^{-1} is of a reasonable magnitude for such an effect. On this reading the observed doublet reflects solvational inequivalence of the ligands rather than the intrinsic symmetry splitting of the isolated complex, a distinction that could be tested by repeating the ground-state calculation with explicit water molecules hydrogen-bonded to the chelate oxygens.

Consequences for the excited-state prediction

The ground-state and excited-state splittings reported here have different physical origins, and the shortfall in the former does not propagate to the latter. The 18 cm^{-1} observed in the ground state is attributed above to solvational inhomogeneity, which the model omits. The 57 cm^{-1} spread predicted for the excited states arises instead from an intrinsic electronic effect, the localisation of charge on one of the three ligands, which the model does describe.

More importantly, the seventeenfold widening between the ground and excited states is an internal comparison: the same functional, basis, solvation model and geometry treatment are applied on both sides, so whatever is absent from the ground-state description is equally absent from the excited-state one and largely cancels in the ratio. The two contributions are moreover additive in sign. An excited state subject to both charge localisation and solvational inhomogeneity should display a multiplet broader than the computed 57 cm^{-1} , not narrower, so the prediction of a substantially widened ν_8 region in the transient spectrum is a conservative one.

2.8 The 941 cm^{-1} transient band

The transient spectra show a band at 941 cm^{-1} , absent from the ground state and blue-shifted by 63 cm^{-1} from the ground-state ν_8 position at 878 cm^{-1} , at both excitation wavelengths. Three readings of this feature can be assessed against the calculations.

The first, that it is ν_8 itself stiffened, is not reproduced. The computed ν_8 descendants shift by -33 to $+24\text{ cm}^{-1}$ in root 15 and -43 to $+11\text{ cm}^{-1}$ in root 61; none approaches $+63\text{ cm}^{-1}$.

The second is that the observed feature corresponds to one of the modes that the calculation does predict to shift strongly to the blue. The largest such shifts are $+97.2\text{ cm}^{-1}$ (root 15, $1040.2 \rightarrow 1137.4\text{ cm}^{-1}$) and $+67.5\text{ cm}^{-1}$ (root 61, $1031.4 \rightarrow 1098.9\text{ cm}^{-1}$). The latter in particular matches the observed $+63\text{ cm}^{-1}$ closely in magnitude. Both parent modes are C–H in-plane bending vibrations of ν_9 character (Table 1), rather than ν_8 . This reading is attractive in that the calculation reproduces a blue shift of the observed size without adjustment; it requires, however, that the transient band be reassigned away from the ν_8 coordinate. It is also consistent with the low Duschinsky overlaps of these transitions, 0.29 and 0.30, which indicate that the blue-shifting excited-state modes are strongly mixed and cannot be traced to a single ground-state parent in any case.

The third possibility is that the ground-state ν_8 splitting is systematically underestimated. The calculation gives a 3.4 cm^{-1} span where experiment resolves 18 cm^{-1} , a factor of five. If the excited-state span is underestimated by a comparable factor, the true multiplet would extend well beyond the computed 57 cm^{-1} and could readily encompass a component 63 cm^{-1} above the ground-state origin.

On the present evidence the second reading is the best supported: the calculation produces excited-state modes blue-shifted by $65\text{--}97\text{ cm}^{-1}$ of the right magnitude, but their mixed character means the transient band is best described as a blue-shifted C–H bending/skeletal combination rather than as ν_8 .

2.9 Vibrational cooling: a density-of-states argument

The experiment finds vibrational cooling after 400 nm excitation to be four to five times faster ($\tau_2 \approx 18\text{--}32\text{ ps}$) than after 270 nm excitation ($\tau_2 \approx 80\text{--}130\text{ ps}$). The properties of the two states considered individually do not account for this ordering, and in isolation would suggest the opposite: root 61 places more than twice the reorganisation energy into the low-frequency skeletal modes that couple most efficiently to a solvent bath ($\sum S_k \approx 15.9$ against 6.4) and drives the Fe–O coordinate 2.4 times harder, both of which would favour faster relaxation, while the Duschinsky mixing metrics differ by only 5%.

The resolution lies not in the properties of either state but in how many states lie beneath it. Table 7 collects the combined sextet and quartet manifolds at the Franck–Condon geometry.

Table 7: Electronic states below and around each excitation, from 80 sextet roots (TDA-TD-DFT) and 140 quartet roots (spin-flip TD-DFT) at the ground-state geometry.

	Root 15 (2.923 eV)	Root 61 (4.517 eV)
Sextet states below	14	60
Quartet states below	17	59
Total states below	31	119
Nearest sextet below / cm^{-1}	1049	605
Nearest quartet / cm^{-1}	250	121
Local quartet density / eV^{-1}	14	26
Local sextet density / eV^{-1}	12	24
Reorganisation energy / eV	0.315	0.589

Root 61 has *four times* as many electronic states below it as root 15, 119 against 31. The measured ratio of cooling times is four to five. We propose that this correspondence is not accidental, and that the measured τ_2 is not the intrinsic solute–solvent energy-transfer time but the duration of the electronic cascade, convolved with cooling.

The argument is as follows. Each internal-conversion or intersystem-crossing step in the cascade converts an electronic energy gap into vibrational quanta. While the cascade is still running it continuously resupplies vibrational energy that the solvent is simultaneously removing, so the vibrational temperature remains elevated for as long as the cascade lasts. For root 15, 2.923 eV is released through some 31 states: the cascade is short, the energy is delivered promptly, and what is subsequently measured approaches the true cooling time. For root 61, 4.517 eV is released through some 119 states; the cascade outlasts the cooling, and the apparent τ_2 is set by the cascade rather than by the bath. This also explains why the same three modes are populated with the same rise behaviour at both wavelengths: they are reporters of a vibrational bath that is being fed from above, consistent with their carrying only 2–6% of the reorganisation energy.

The quartet manifold supplies a shortcut, and it is decisive for root 15 in a way it is not for root 61. Root 15 lies in a sextet-sparse region, 1049 cm^{-1} from the nearest same-spin state, but only 250 cm^{-1} (31 meV) from a quartet. That gap is comparable to $k_B T$ at room temperature (207 cm^{-1}) and small against the Fe(III) spin–orbit coupling constant ($\zeta \approx 400 \text{ cm}^{-1}$), so the crossing is close to resonant. Same-spin internal conversion is slow in this region and intersystem crossing should dominate. Crucially, the quartet manifold extends down to 1.395 eV, below the lowest sextet at 1.94 eV: a single spin-flip step transfers the system onto a ladder that reaches further down than the sextet manifold does, bypassing much of it.

Root 61 is coupled to the quartets even more strongly, the nearest lying only 121 cm^{-1} away, but crossing gains it nothing: at 4.517 eV it still has 59 quartets beneath it. The shortcut is only useful to a state already near the bottom of the manifold. Two further observations reinforce this. Root 61’s gaps are roughly half the size of root 15’s (605 against 1049 cm^{-1} for the nearest sextet), so by the energy-gap law each individual step is faster — but each also releases less energy, so more steps are required to dissipate more total energy, giving a finer and slower staircase. And root 61 drives the Fe–O coordinate 2.4 times harder, which should promote spin–orbit crossing more effectively per event; that it nonetheless relaxes more slowly is itself evidence that the bottleneck is the length of the cascade rather than the rate of any individual step.

We emphasise the status of this argument. It is a statistical, density-of-states explanation resting on four mutually consistent quantities — state counts, gap sizes, quartet proximity and reorganisation energies — and its strongest single element is the agreement between the fourfold ratio of state counts and the fourfold to fivefold ratio of measured cooling times. It is not a rate calculation: no spin–orbit matrix elements, Fermi-golden-rule rates or crossing geometries have been computed. It also rests entirely on TD-DFT state densities. Individual TD-DFT excitation

energies for a complex of this kind carry uncertainties of a few tenths of an electronvolt, and spin-flip TD-DFT shows some spin contamination here ($\langle S^2 \rangle$ up to 4.93 against 3.75). Counting arguments are more forgiving than individual energies, since errors partly cancel when states are enumerated rather than placed individually, but the absolute counts remain dependent on the functional and on the number of roots requested. The proposal should therefore be read as a plausible and internally consistent mechanism rather than a demonstrated one. A quantitative test would require spin-orbit couplings $\langle {}^6\Gamma | \hat{H}_{\text{SO}} | {}^4\Gamma \rangle$ between the relevant pairs, from which Fermi-golden-rule rates could be evaluated with the reorganisation energies reported here.

2.10 Summary

1. The sextet ground state is a genuine minimum (123 real modes) at TPSSh/SMD(water); the Franck-Condon projection is complete to better than 0.21%.
2. Huang-Rhys factors reach $S_k \approx 5$ and the total reorganisation energies are 0.315 and 0.589 eV. Structural reorganisation is substantial, not negligible.
3. That reorganisation lies almost entirely outside the monitored bands, at 150–450 cm^{-1} and 1559 cm^{-1} . The observed modes report the vibrational bath, not the primary distortion.
4. The two excitation windows are distinguished quantitatively: root 15 drives the ligand C=C/C=O stretch, root 61 the Fe-O framework two to three times more strongly.
5. The 1442–1446 cm^{-1} cluster is methyl deformation (79% hydrogen); the conjugated C=C/C=O stretch lies at 1559 cm^{-1} , and the 811 cm^{-1} triplet is a C-H wag distinct from ν_8 .
6. Excited-state normal modes are mixtures of four to five ground-state modes; ground-state labels should not be transferred to transient bands without quoting the Duschinsky overlap.
7. A charge-transfer excitation in a tris-chelate must lift the degeneracy of the doubly degenerate E ligand vibrations. A hole shared between three equivalent rings localises whenever vibronic coupling exceeds the intrinsic splitting of the manifolds, which here is only 1.3 cm^{-1} , and the localising distortion destroys the equivalence the degeneracy rests on. Neither state is orbitally degenerate, so the mechanism is second-order rather than first-order Jahn-Teller.
8. The prediction is borne out structurally. The six Fe-O distances span 0.014 Å in the ground state but 0.098 and 0.241 Å in the two excited states, root 15 showing two short and four long distances, the signature of a hole localised on one chelate ring.
9. It is borne out vibrationally, and globally: all twelve near-degenerate ground-state clusters widen in both excited states without exception, by a median factor of 13 and 15 and by no less than three. The ν_8 manifold accessible to experiment widens by 16.7 and 16.6, close to the medians — observable rather than exceptional.
10. The observed 941 cm^{-1} transient band is best assigned to a blue-shifted, strongly mixed C-H bending/skeletal combination rather than to a stiffened ν_8 .
11. The Fe-O stretch is reproduced to 14 cm^{-1} . The ground-state ν_8 splitting is underestimated (1.3–3.4 cm^{-1} computed against 18 cm^{-1} observed), most probably because a continuum solvation model preserves the equivalence of the three ligands that instantaneous hydrogen bonding breaks. The predicted excited-state widening is unaffected, being an internal comparison at a common level of theory and additive with any solvational contribution.
12. Root 61 has four times as many electronic states below it as root 15 (119 against 31, counting both manifolds), matching the fourfold to fivefold ratio of measured cooling times. We propose that τ_2 measures the duration of the electronic cascade rather than the intrinsic cooling time.

13. A near-resonant quartet 250 cm^{-1} from root 15, in a region where same-spin internal conversion is slow, gives the 400 nm excitation a spin-flip shortcut onto a manifold reaching below the lowest sextet. Root 61 couples to the quartets more strongly still but gains nothing by it, having 59 quartets beneath it.